Hierarchical models to predict predation in complex trophic communities

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Trophic interactions make the backbone of ecological communities. From herbivory to predation, these interactions of varying strength shape ecosystem and their related functions. There is a pressing need to be able to predict interaction strength at the scale of whole communities. While functional responses focus at the pairwise interaction level, we question here if the same underlying mechanisms could be leveraged to make predictions at the scale of the community. In this study, we develop 4 hierarchical models of increasing complexity based on the type I and the type II functional responses to predict biomass fluxes between consumers and resources at the community level. The main objective is to determine if simple mechanism such as mass action (type I) and the saturation of fluxes (type II) are sufficient to predict biomass fluxes. To do so, we collected 19 Ecopath networks encompassing 1380 interactions across marine, aquatic and terrestrial ecosystems. While we can not reject the mechanism of saturating fluxes (type II), our results suggest that a mass action mechanism is sufficient and more robust, in the context of our data, to predict biomass fluxes between consumers and resources at the community level. We also recovered the expected allometric relationships between consumers body mass and their space clearance rate parameters within the mass action model, further solidifying our findings. Further modelisation is needed, specifically a trait-matching model with more data, to develop a model generalizable to more taxons and types of trophic interactions.

Keywords:
pairwise trophic interactions
ecological networks
biomass flows
mass action
space clearance rate
bayesian analysis

Trophic interactions make the backbone of ecological communities (Bellingeri & Bodini 2016; DeLong 2021; Wootton & Emmerson 2005), linking every component through nutrient and energy flows (Lindeman 1942). Consumer-resource interactions of varying strength play a crucial role in shaping important ecosystem functions and processes (Curtsdotter *et al.* 2019; Duffy 2002; Koltz *et al.* 2018; Montoya *et al.* 2003; Wilmers *et al.* 2012) such as disease dynamics, invasive species and biogeochemical cycles (Estes *et al.* 2011). Identifying species interactions and quantifying their strength is necessary to understand community composition (Paine 1992), stability (Bascompte *et al.* 2005; de Ruiter *et al.* 1995; Neutel 2002; Nilsson & McCann 2016) and the resulting associated ecosystem services (Montoya *et al.* 2003). Global changes have a diversity of impacts on species themselves, but also how they interact and the resulting assemblages and functioning (Koltz *et al.* 2018; Lurgi *et al.* 2012). Empirical observation of these interactions and the evaluation of their strength is however challenging, if not unfeasable due to their sheer amount, sampling biases and the emergence of novel interactions between species that never co-occurred (Aguiar *et al.* 2019; Jordano 2016; Wootton & Emmerson 2005). The study of ecological networks has therefore recently moved to the development of predictive tools to overcome the limitations associated with empirical sampling of trophic interactions (Poisot *et al.* 2016). Different models are derived from food web theory, such as the niche model (Williams & Martinez 2000), the nested-hierarchy model (Cattin *et al.* 2004), trait-matching model (Bartomeus *et al.* 2016) or physical constraints (Portalier *et al.* 2019). More recently, Strydom *et al.* (2021) developed a general framework to predict interaction probabilities between species in a community context. These methods are however mostly qualitative, i.e. they are meant to predict if there is an interaction or not, and there is a pressing need to expand them to quantitative interactions, where the strength of interaction between any two organisms is described.

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There are many definitions of interaction strength (Berlow *et al.* 2004), and among them biomass flows can be particularly suited for the development of predictive models. Biomass flows between consumers and resources can be derived from functional traits, with the promise of a standardized measure independent of taxa involved (Berlow *et al.* 2004). Quantitative predictive models of biomass flows might also offer potential support to the study of biodiversity and ecosystem functioning relationship, as it seeks to depict all energy flows within communities (Barnes *et al.* 2018). Biomass flows are defined as a quantity of resource biomass that is consumed by a consumer per unit area and unit of time, reporting how energy circulates through the food web (Lindeman 1942). Interaction strengths between consumers and resources is closely related to predation rate in a food web model, namely the functional responses](Rall *et al.* 2012). Functional response theory aims at describing the change in feeding rate of a consumer, or how many prey or resource a consumer can predate per unit area over time, relative to prey or resource availability (Holling 1959a, b; Solomon 1949). Where the functional response describes the actual number of individual prey or resource predated by a consumer per unit area over time, biomass flows describes the amount of biomass flowing from a prey or resource to a consumer per unit area over time. Since both quantities report the rate of the same underlying process that is predation, we anchored the development of our modeling approach in functional response theory.

A fundamental principle underlying the majority of functional responses is the mass action, stating that predator and prey move randomly in an area and encounter at a rate proportional to the product of their abundance and their velocity. Here we will focus on the two most commonly used types of functional responses. The type I functional response describes mass action only, where the amount of resource consumed depends on the resource and consumer densities, and the rate (formerly reffered to as attack rate and hereafter space clearance rate (DeLong 2021)) at which a consumer searches its home range for resources, leading to random encounters followed by detection, attack and consumption. The type II functional response describes the same process as the type I, but with the addition that consumption saturates with prey availability because of the time required to process prey. These processes are influenced by a multitude of functional traits and environmental conditions (Beardsell *et al.* 2021, 2022). The space clearance rate parameter varies with consumers and resources intrinsic characteristics such as absolute and relative velocities, perception of the environment and cognitive abilities (DeLong 2021). It is also influenced by externalities such as habitat complexity (Barrios-O'Neill *et al.* 2016; Hall *et al.* 2000; Manatunge *et al.* 2000) and the presence of refugia (Hoddle 2003) impeding detection probability. The dimensionality of an interaction, whether it takes place in a two or three dimensional plane, also impacts encounter rates and detection probability (Pawar *et al.* 2012, 2019). For instance, a predatory bird searching for preys from above or fishes grazing the benthic surface are examples of 2D planes while aerial insectivorous birds pursuing flying insects or fishes chasing preys in the water column would be in a 3D plane. Functional response theory therefore makes a good start to develop predictive models of quantitative interactions since widely measured functional traits can be combined with consumer and prey density to predict biomass flows.

The space clearance rate, in units of space per time, reports the amount of area search by the predator in a time interval. As with many biological processes, the space clearance rate was shown to vary with consumers body mass. Indeed, multiple studies, summarized in the FoRAGE database (Uiterwaal *et al.* 2022), showed an allometric scaling between the space clearance rate and the body mass of consumers. This observation is coherent with (Hirt *et al.* 2017, 2020) who documented that the maximum velocity of an organism scales with its body mass. Handling time is also expected to follow the same scaling relationship (DeLong 2021), although recent studies are now suggesting that consumers handling time would not be as important at natural resources densities (Beardsell *et al.* 2021; Chan *et al.* 2017; Coblentz *et al.* 2023; Preston *et al.* 2018). Ultimately, an accurate evaluation of these parameters is necessary as they are widely used in theoretical models (Brose *et al.* 2006; Brose *et al.* 2008; Pawar *et al.* 2012). Restraining the parameter space could greatly bonify studies of species coexistence, community stability and trophic regulation. It could also help the parameterization of global ecosystem models used for biodiversity scenarios (Harfoot *et al.* 2014}). Following (Yodzis & Innes 1992), there are a few meta-analyses looking at the scaling of feeding rates with body-mass (Rall *et al.* 2012). These are however concentrated on a re-analysis of univariate functional responses, mostly from experimental studies, and there is currently no study aiming at reconstructing quantitative interaction networks.

Our main objective is to test if we could model quantitative interactions in complex food webs with easily accessible information such as abundance and body mass. We compared statistical models derived from functional response theory, including type I and type II forms. We also investigate if allometric relationships between space clearance rate, handling time and consumer body mass can contribute to the prediction of biomass flows in diverse food webs. To do so, we developed four hierarchical models of pairwise quantitative interactions of increasing complexity and evaluated them with empirical data of biomass flows between consumers and resources. We curated Ecopath networks (Christensen *et al.* 2005) describing marine, aquatic

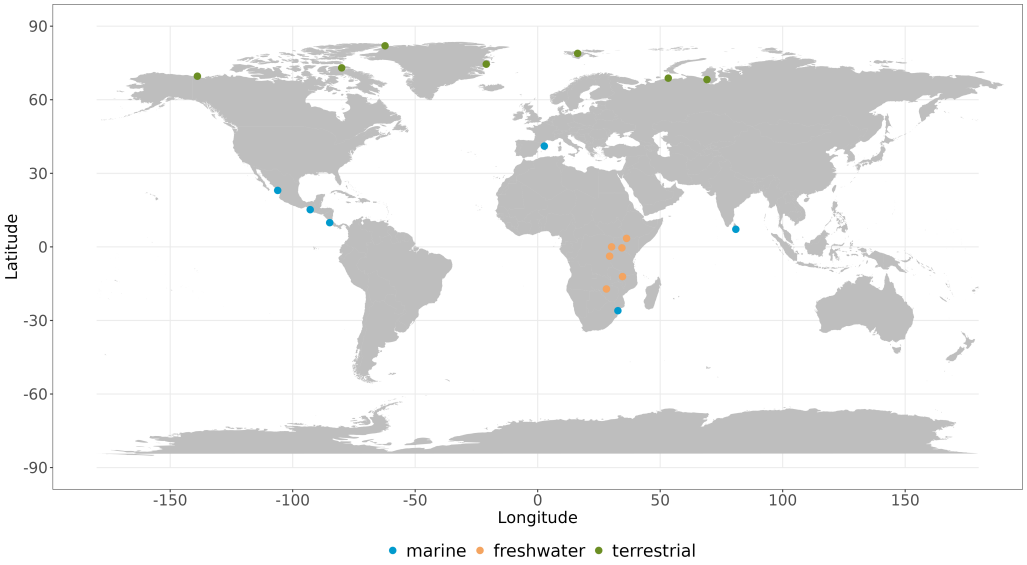

Figure 1 Map of Ecopath models locations used in the present study.

and terrestrial food webs and extracted 1380 pairwise trophic interactions between consumers and resources. Each model represents its own hypothesis and their comparison allow the evaluation of different mechanisms driving interactions.

1 Material and methods

A litterature search was performed to gather empirically sampled trophic networks with measurements of biomass flows from resources to consumers, along with population densities. Food web data are notoriously heterogeneous (Mestre *et al.* 2022) and we therefore relied on Ecopath models (Christensen *et al.* 2005) to normalize data collection. Ecopath is a modelling software that relies on a mass-balance premiss and a multitude of parameters (production/biomass ratio, consumption/biomass ratio, diet compositions etc.) to provide a realistic static image of the flows of biomass between resources and consumers in an ecosystem (Christensen *et al.* 2005). It was shown the Ecopath methodology reduces biases in comparative analyses among food webs (Brimacombe, in review). We therefore collected openly available models from Ecobase (Colléter *et al.* 2013) and worked with a subset of the models present in (Jacquet *et al.* 2016). Taxonomic resolution of Ecopath models is highly variable and it is not uncommon that species are lumped into functional groups. The models were therefore selected based on the taxonomic resolution with the criteria that most nodes must be resolved at the species level, or that species within functional groups were taxonomically resolved. We thus curated 19 Ecopath models (detailed list in supplementary material) from aquatic, marine and terrestrial environments (fig. 1), and trophic guilds spanning from mammal carnivores and herbivores, birds, pelagic carnnivorous and herbivorous fish, demersal carnivorous and herbivorous fish and invertebrates.

Species identity in the Ecopath models were validated with the original publication and estimates of adult mean body mass was retrived from literature. Body mass for terrestrial species came from different sources such as the GATEWAy database (Brose 2018), the original publication for which the Ecopath model was developped or grey literature. Body mass for marine and freshwater species came from Fishbase (Froese *et al.* 2023) and Sealifebase (Palomares *et al.* 2023). When only body length was available, Fishbase and Sealifebase calculated weight from body length using a bayesian length-weight model. The midpoint between the minimum and maximum measurements was taken when a mean was not available. Density (number of individuals per unit surface, N/km^2) was computed by dividing total biomass per unit surface B (metric tons/km^2) by mean body mass M (metric tons). The 19 Ecopath models were aggregated in an edge list with quantitative biomass flows (metric tons/km^2/year), for a total of 1380 interactions between

154 consumer species and 153 resource species.

1.1. Model description We developed four different hierarchical models of increasing complexity based on the type I and type II functional responses, and a null model. Models were then parametrized with Hamiltonian Monte Carlo (Monnahan *et al.* 2017), compared and ranked from better to worst based on their prediction accuracy. The functional form of these relationships are presented below.

1.1.1 Model 0 - Null model The null model acts as a basis of comparison with the other models and does not portray any ecological realism of a trophic interaction. Model 0 hypothesizes that the flow of biomass F_{ij} from a prey i to its predator j is constant with average K, irrespective of predator identity:

$$F_{ij} = K$$

1.1.2 Model 3 - Single-species saturating model Model 3 keeps the assumptions of model 2 with the addition of handling time, thereby representing a single species Type II functional response. The space clearance rate parameter α_j and the handling time h_j are both consumer-specific:

$$F_{ij} = \frac{\alpha_j B_i N_j}{1 + h_j \alpha_j B_i N_j}$$

References

Aguiar, M.A.M. de, Newman, E.A., Pires, M.M., Yeakel, J.D., Boettiger, C., Burkle, L.A., *et al.* (2019). Revealing biases in the sampling of ecological interaction networks. *PeerJ*, 7, e7566.

Barnes, A.D., Jochum, M., Lefcheck, J.S., Eisenhauer, N., Scherber, C., O'Connor, M.I., *et al.* (2018). Energy Flux: The Link between Multitrophic Biodiversity and Ecosystem Functioning. *Trends in Ecology & Evolution*, 33, 186–197.

Barrios-O'Neill, D., Kelly, R., Dick, J.T.A., Ricciardi, A., MacIsaac, H.J. & Emmerson, M.C. (2016). On the context-dependent scaling of consumer feeding rates. *Ecology Letters*, 19, 668–678.

Bartomeus, I., Gravel, D., Tylianakis, J.M., Aizen, M.A., Dickie, I.A. & Bernard-Verdier, M. (2016). A common framework for identifying linkage rules across different types of interactions. *Functional Ecology*, 30, 1894–1903.

Bascompte, J., Melian, C.J. & Sala, E. (2005). Interaction strength combinations and the overfishing of a marine food web. *Proceedings of the National Academy of Sciences*, 102, 5443–5447.

Beardsell, A., Gravel, D., Berteaux, D., Gauthier, G., Clermont, J., Careau, V., *et al.* (2021). Derivation of Predator Functional Responses Using a Mechanistic Approach in a Natural System. *Frontiers in Ecology and Evolution*, 9.

Beardsell, A., Gravel, D., Clermont, J., Berteaux, D., Gauthier, G. & Bêty, J. (2022). A mechanistic model of functional response provides new insights into indirect interactions among arctic tundra prey. *Ecology*, 103.

Bellingeri, M. & Bodini, A. (2016). Food web's backbones and energy delivery in ecosystems. *Oikos*, 125, 586–594.

Berlow, E.L., Neutel, A.-M., Cohen, J.E., de Ruiter, P.C., Ebenman, B., Emmerson, M., *et al.* (2004). Interaction strengths in food webs: Issues and opportunities. *Journal of Animal Ecology*, 73, 585–598.

Brose, U. (2018). GlobAL daTabasE of traits and food Web Architecture (GATEWAy) version 1.0.

Brose, U., Ehnes, R.B., Rall, B.C., Vucic-Pestic, O., Berlow, E.L. & Scheu, S. (2008). Foraging theory predicts predator–prey energy fluxes. *Journal of Animal Ecology*, 77, 1072–1078.

Brose, U., Williams, R.J. & Martinez, N.D. (2006). Allometric scaling enhances stability in complex food webs. *Ecology Letters*, 9, 1228–1236.

Cattin, M.-F., Bersier, L.-F., Banašek-Richter, C., Baltensperger, R. & Gabriel, J.-P. (2004). Phylogenetic constraints and adaptation explain food-web structure. *Nature*, 427, 835–839.

Chan, K., Boutin, S., Hossie, T.J., Krebs, C.J., O'Donoghue, M. & Murray, D.L. (2017). Improving the assessment of predator functional responses by considering alternate prey and predator interactions. *Ecology*, 98, 1787–1796.

Christensen, V., Walters, C. & Pauly, D. (2005). Ecopath with Ecosim: A User's Guide. *Fisheries Centre, University of British Columbia, Vancouver, Canada and ICLARM, Penang, Malaysia*, 12.

Coblentz, K.E., Novak, M. & DeLong, J.P. (2023). Predator feeding rates may often be unsaturated under typical prey densities. *Ecology Letters*, 26, 302–312.

Colléter, M., Valls, A., Guitton, J., Lyne, M., Sánchez, F.A.-., Christensen, V., *et al.* (2013). *EcoBase: A repository solution to gather and communicate information from EwE models* (Research Report No. 21). Fisheries Centre, University of British Columbia, Canada.

Curtsdotter, A., Banks, H.T., Banks, J.E., Jonsson, M., Jonsson, T., Laubmeier, A.N., *et al.* (2019). Ecosystem function in predator–prey food webs—confronting dynamic models with empirical data. *Journal of Animal Ecology*, 88, 196–210.

de Ruiter, P.C., Neutel, A.-M. & Moore, J.C. (1995). Energetics, Patterns of Interaction Strengths, and Stability in Real Ecosystems. *Science*, 269, 1257–1260.

DeLong, J.P. (2021). *Predator Ecology: Evolutionary Ecology of the Functional Response*. Oxford University Press.

Duffy, J.E. (2002). Biodiversity and ecosystem function: The consumer connection. *Oikos*, 99, 201–219.

Estes, J.A., Terborgh, J., Brashares, J.S., Power, M.E., Berger, J., Bond, W.J., *et al.* (2011). Trophic Downgrading of Planet Earth. *Science*, 333, 301–306.

Froese, R., Pauly, D. & Editors. (2023). FishBase.

Hall, R.O., Wallace, J.B. & Eggert, S.L. (2000). Organic matter flow in stream food webs with reduced detrital resource base. *Ecology*, 81, 3445–3463.

Harfoot, M.B.J., Newbold, T., Tittensor, D.P., Emmott, S., Hutton, J., Lyutsarev, V., *et al.* (2014). Emergent Global Patterns of Ecosystem Structure and Function from a Mechanistic General Ecosystem Model. *PLOS Biology*, 12, e1001841.

Hirt, M.R., Jetz, W., Rall, B.C. & Brose, U. (2017). A general scaling law reveals why the largest animals are not the fastest. *Nature Ecology & Evolution*, 1, 1116–1122.

Hirt, M.R., Tucker, M., Müller, T., Rosenbaum, B. & Brose, U. (2020). Rethinking trophic niches: Speed and body mass colimit prey space of mammalian predators. *Ecology and Evolution*, 10, 7094–7105.

Hoddle, M.S. (2003). The effect of prey species and environmental complexity on the functional response of Franklinothrips orizabensis: A test of the fractal foraging model. *Ecological Entomology*, 28, 309–318.

Holling, C.S. (1959a). Some Characteristics of Simple Types of Predation and Parasitism. *The Canadian Entomologist*, 91, 385–398.

Holling, C.S. (1959b). The Components of Predation as Revealed by a Study of Small-Mammal Predation of the European Pine Sawfly. *The Canadian Entomologist*, 91, 293–320.

Jacquet, C., Moritz, C., Morissette, L., Legagneux, P., Massol, F., Archambault, P., *et al.* (2016). No complexity–stability relationship in empirical ecosystems. *Nature Communications*, 7, 12573.

Jordano, P. (2016). Chasing Ecological Interactions. *PLOS Biology*, 14, e1002559.

Koltz, A.M., Classen, A.T. & Wright, J.P. (2018). Warming reverses top-down effects of predators on belowground ecosystem function in Arctic tundra. *Proceedings of the National Academy of Sciences*, 115, E7541–E7549.

Lindeman, R.L. (1942). The Trophic-Dynamic Aspect of Ecology. *Ecology*, 23, 399–417.

Lurgi, M., López, B.C. & Montoya, J.M. (2012). Novel communities from climate change. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 367, 2913–2922.

Manatunge, J., Asaeda, T. & Priyadarshana, T. (2000). The Influence of Structural Complexity on Fish–zooplankton Interactions: A Study Using Artificial Submerged Macrophytes. *Environmental Biology of Fishes*, 58, 425–438.

Mestre, F., Rozenfeld, A. & Araújo, M.B. (2022). Human disturbances affect the topology of food webs. *Ecology Letters*, 25, 2476–2488.

Monnahan, C.C., Thorson, J.T. & Branch, T.A. (2017). Faster estimation of Bayesian models in ecology using Hamiltonian Monte Carlo. *Methods in Ecology and Evolution*, 8, 339–348.

Montoya, J.M., Rodríguez, M.A. & Hawkins, B.A. (2003). Food web complexity and higher-level ecosystem services. *Ecology Letters*, 6, 587–593.

Neutel, A.-M. (2002). Stability in Real Food Webs: Weak Links in Long Loops. *Science*, 296, 1120–1123.

Nilsson, K.A. & McCann, K.S. (2016). Interaction strength revisited—clarifying the role of energy flux for food web stability. *Theoretical Ecology*, 9, 59–71.

Paine, R.T. (1992). Food-web analysis through field measurement of per capita interaction strength. *Nature*, 355, 73–75.

Palomares, M.L.D., Pauly, D. & Editors. (2023). SeaLifeBase.

Pawar, S., Dell, A.I., Lin, T., Wieczynski, D.J. & Savage, V.M. (2019). Interaction Dimensionality Scales Up to Generate Bimodal Consumer-Resource Size-Ratio Distributions in Ecological Communities. *Frontiers in Ecology and Evolution*, 7, 202.

Pawar, S., Dell, A.I. & Van M. Savage. (2012). Dimensionality of consumer search space drives trophic interaction strengths. *Nature*, 486, 485–489.

Poisot, T., Stouffer, D.B. & Kéfi, S. (2016). Describe, understand and predict: Why do we need networks in ecology? *Functional Ecology*, 30, 1878–1882.

Portalier, S.M.J., Fussmann, G.F., Loreau, M. & Cherif, M. (2019). The mechanics of predator–prey interactions: First principles of physics predict predator–prey size ratios. *Functional Ecology*, 33, 323–334.

Preston, D.L., Henderson, J.S., Falke, L.P., Segui, L.M., Layden, T.J. & Novak, M. (2018). What drives interaction strengths in complex food webs? A test with feeding rates of a generalist stream predator. *Ecology*, 99, 1591–1601.

Rall, B.C., Brose, U., Hartvig, M., Kalinkat, G., Schwarzmüller, F., Vucic-Pestic, O., *et al.* (2012). Universal temperature and body-mass scaling of feeding rates. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 367, 2923–2934.

Solomon, M.E. (1949). The Natural Control of Animal Populations. *Journal of Animal Ecology*, 18, 1–35.

Strydom, T., Catchen, M.D., Banville, F., Caron, D., Dansereau, G., Desjardins-Proulx, P., *et al.* (2021). A roadmap towards predicting species interaction networks (across space and time). *Philosophical Transactions of the Royal Society B: Biological Sciences*, 376, 20210063.

Uiterwaal, S.F., Lagerstrom, I.T., Lyon, S.R. & DeLong, J.P. (2022). FoRAGE database: A compilation of functional responses for consumers and parasitoids. *Ecology*, 103, e3706.

Williams, R.J. & Martinez, N.D. (2000). Simple rules yield complex food webs. *Nature*, 404, 180–183.

Wilmers, C.C., Estes, J.A., Edwards, M., Laidre, K.L. & Konar, B. (2012). Do trophic cascades affect the storage and flux of atmospheric carbon? An analysis of sea otters and kelp forests. *Frontiers in Ecology and the Environment*, 10, 409–415.

Wootton, J.T. & Emmerson, M. (2005). Measurement of Interaction Strength in Nature. *Annual Review of Ecology, Evolution, and Systematics*, 36, 419–444.

Yodzis, P. & Innes, S. (1992). Body Size and Consumer-Resource Dynamics. *The American Naturalist*, 139, 1151–1175.